7.4 Solving Polynomial Equations in Factored Form

Name _____ Period ____ Date _____

Algebra I 7.4 (30-minute plan) · Student sheet

I can…

- use the Zero-Product Property to solve equations.
- factor out the greatest common monomial factor of a polynomial.
- solve polynomial equations by factoring out the GCF.

Vocabulary · fill in the missing word

Factored form — a _____ written as a product of factors, like $(x - 4)(x + 1)$

Zero-Product Property — if $ab =$ _____, then $a = 0$ or $b = 0$

Roots — the solutions of a _____ equation

Repeated roots — a root that comes from a factor that _____ more than once

Part 1 · Notes and examples (fill in each note, then work the example)

Zero-Product Property: if $ab = 0$, then _____ or _____.

Solving a factored equation: To solve a factored equation, set _____ factor equal to 0.

Example 1

Solve the equation.

$(x - 4)(x + 1) = 0$

Repeated roots: A root that comes from a factor that appears more than once is a _____ root.

Example 2

Solve the equation.

$(n - 2)^2(n + 7) = 0$

Solving by factoring: To solve by factoring: get _____ on one side, _____, then use the Zero-Product Property.

Careful: Don’t divide both sides by x: you lose the solution _____.

Example 3

Solve the equation.

$4x^2 = 20x$

Part 2 · Partner practice

Partner practice

Solve the equation.

$(2x - 7)(x + 3) = 0$